

Adverse Drug Reactions and Pharmacovigilance

Theory exam concepts + CDSCO ADR reporting practical — pharmacology lecture notes

Learning Objectives

- Define an adverse drug reaction (ADR)
- Classify the types of ADRs
- Define pharmacovigilance
- Describe how to report an ADR

Historical Perspective: The Thalidomide Catastrophe

In the late 1950s, thalidomide was marketed as a treatment for nausea and vomiting in pregnancy. Babies born to women who took it developed severe limb deformities — very short limbs resembling a seal's flippers, a condition called **phocomelia**. This catastrophe triggered the start of formal adverse reaction monitoring worldwide.

Therapeutic Effect vs. Adverse Drug Reaction

- **Therapeutic effect** — the intended, expected response to a drug
- **Adverse drug reaction (ADR)** — any undesirable, unintended effect of a drug
- **Side effect (SE)** — a relatively harmless undesirable effect
- **Toxic effect** — a harmful undesirable effect

Types of Toxic (Adverse) Drug Effects

Type	Mechanism	Example
Pharmacological toxicity	Exaggerated / increased response to the drug	Nifedipine (Ca ²⁺ channel blocker) → excess vasodilation → hypotension
Pathological toxicity	Toxic metabolite accumulates when normal clearance pathway is overwhelmed	Paracetamol (APAP) overdose → liver failure when glutathione is depleted
Genotoxicity	Damage to genetic material	Anticancer drugs
Allergic reaction	Immune-mediated; needs prior sensitization	Penicillins (see hypersensitivity types below)
Idiosyncratic reaction	Unusually high or low sensitivity to a drug	—
Pharmacogenetic variation	Genetic variation alters metabolizing enzymes or transporters	Psoriasis → hemolysis in G6PD-deficient patients
Teratogenicity	Drug crosses placenta and disrupts fetal development	Thalidomide (category X)

Allergic (Hypersensitivity) Reactions

Occur only in a person previously exposed/sensitized to the drug. Four types:

Type	Also known as	Mediator	Mechanism / example
Type I	Immediate hypersensitivity	IgE	Allergen binds IgE on mast cells → degranulation → urticaria, bronchoc
Type II	Cytolytic reaction	IgG, IgM + complement	Lysis of blood cells. Example: penicillin-induced hemolytic anemia
Type III	Arthus reaction	IgG + complement	Immune complex deposits in vascular endothelium → serum sickness (s
Type IV	Delayed hypersensitivity	Sensitized macrophages and T-lymphocytes	Cell-mediated inflammatory response. Example: lamotrigine

Teratogenicity and Drug Use in Pregnancy

- Drugs taken in the first 17 days after conception (fertilization + implantation) → may cause an unnoticed miscarriage
- **Most vulnerable period: day 18–55** — the period of organogenesis; teratogenic drugs here can cause congenital anomalies
- After day 56 → risk shifts to developmental or functional disorders rather than structural anomalies

Pregnancy Risk Categories

Category	Definition	Example
A	Safest in pregnancy	Levothyroxine
B	No risk in animal studies (no human data), OR animal studies show no risk but human studies show none	Amoxicillin
C	Animal studies show risk but no human data, or data lacking in both	Fluconazole
D	Demonstrated fetal risk, but benefit may outweigh risk	Phenytoin (fetal hydantoin syndrome; surgically correct
X	Must not be used under any circumstances	Thalidomide

Classification by Time of Onset

Onset	Timing
Acute	Within 60 minutes of drug administration
Subacute	60 minutes to 24 hours
Latent	Usually after 2 days

Classification by Severity

Severity	Definition
Minor	No treatment/antidote needed; no prolonged hospitalization
Moderate	Requires a change in therapy or treatment; hospitalization prolonged by at least 1 day
Severe	Life-threatening; produces permanent damage; requires intensive treatment
Lethal	Drug directly or indirectly contributes to the patient's death

Pharmacovigilance

Pharmaco = pertaining to drugs; **vigilance** = to watch. Pharmacovigilance is the science and activities relating to the detection, assessment, understanding, and prevention of adverse drug reactions.

Key Terms

- **Adverse event** — any untoward event during treatment, whether or not related to the drug
- **Serious adverse event** — life-threatening, or results in death, disability, congenital anomaly, hospitalization/prolonged hospitalization, or requires intervention to prevent permanent disability
- **Causality assessment** — determining whether the drug is actually responsible for the ADR

Bradford Hill Criteria (for Causality Assessment)

- **Temporal relationship** — did the reaction occur after the drug was taken?
- **Plausibility** — is there a scientific/theoretical basis for the reaction?
- **Dose–response relationship** — does the reaction increase/decrease with dose?
- **Consistency** — is the reaction reproduced on repeated administration?

Common causality scales: **WHO-UMC scale** and the **Naranjo algorithm**.

Pharmacovigilance Programme of India (PVPI)

- **Goal:** ensure drug use carries more benefit than risk; safeguard the health of the Indian population
- **Objectives:** monitor ADRs, create awareness among healthcare professionals, understand the risk–benefit profile of pharmaceuticals, and issue safety recommendations
- **Who can report:** both healthcare professionals and consumers
- **How to report:** the ADR PvPI mobile app (Google Play Store), or the suspected ADR form from the CDSCO website

Structure of the CDSCO Suspected ADR Reporting Form

(This is the key practical-exam exercise.)

Section	What to enter
1. Patient information	Patient initials (not the name — confidentiality), age or date of birth, gender, weight
2. Suspected adverse reaction	Date reaction started, date recovery began, description of the reaction
3. Suspected medication(s)	Generic/brand name, manufacturer, batch/lot number, expiry date, dose, route, frequency, date started
4. Action taken	Usually drug withdrawn or dose reduced — tick against the relevant drug column
5. Rechallenge	Did the reaction reappear on reintroducing the drug? (Not usually done if the ADR was serious/lethal)
6. Concomitant medical products	OTC medications, dietary supplements etc., with dose, route, frequency, dates, indication
7. Relevant history	Relevant laboratory and medical history
8. Seriousness	Life-threatening? Hospitalization/prolonged hospitalization? Death? Congenital anomaly? Intervention required?
9. Outcome	Recovering / not recovered / completely recovered, etc.

Reporting Pathway

Completed form → nearest **ADR Monitoring Centre (AMC)** → forwarded to the **National Coordination Centre, Ghaziabad, New Delhi**.

Summary

- An ADR is any unintended effect occurring with drug treatment
- ADRs are classified by mechanism (pharmacological, pathological, genotoxic, allergic, idiosyncratic, pharmacogenetic, teratogenic), by onset (acute/subacute/latent), and by severity (minor/moderate/severe/lethal)
- Pharmacovigilance is the monitoring of ADRs
- Report via the ADR PvPI app or the CDSCO suspected-ADR form, routed through the nearest AMC to Ghaziabad

Exam note: Pharmacovigilance concepts (theory) and the CDSCO ADR form (practical exercise) are both important — this topic appears in both exam components.